Lecture Notes:
Math & Statistics for Data Science
Unit 1, Mathematical Fundamentals

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1 Logic

Learning Goals

- Understand what a statement is in mathematical logic.
- Construct truth tables for logical operations.
- Apply De Morgan's laws and implications in reasoning.

Content

- **Statements:** Sentences that have a definite truth value (true/false).
  - Examples: "I washed the car"; $x = 1$.
  - Non-examples: questions, commands.
- **Symbolic representation:** use letters such as $p =$ "I washed the car."
- **Logical operations:**
  - Negation ($\neg p$)
  - Conjunction ($p \wedge q$)
  - Disjunction ($p \vee q$), inclusive vs. exclusive
  - Implication ($p \to q$)

  - Bi-implication ($p \leftrightarrow q$)
  - Converse, inverse, contrapositive

- **Truth tables:** Introduce systematically.
- **Quantifiers:** Universal ($\forall$), Existential ($\exists$), and their negations.

Exercises

- Identify which sentences are statements.
- Build truth tables for given propositions.
- Apply De Morgan's laws to symbolic expressions.

2 Sets

Learning Goals

- Define and use set notation.
- Perform standard set operations.
- Distinguish between subsets, proper subsets, and disjoint sets.

Content

- **Sets:** collections of objects with a defining property.
  - Elements, membership ($\in$).
  - Examples: $\{1, 2, 3\}$, singleton sets, empty set.
  - Rule-based definitions: $\{x \mid x > 0\}$.
- **Subsets:** $A \subseteq B$, including proper subsets and the empty set.
- **Special sets:** $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, etc.
- **Intervals:** open, closed, half-open.
- **Operations:**

- Union: $A \cup B$
- Intersection: $A \cap B$
- Difference: $A \setminus B$
- Disjoint sets
- Indexed unions/intersections

- **Tuples and ordered pairs:** introduction to vectors.

Exercises

- Classify given sets as subsets of others.
- Interval "surgery" problems (finding overlaps/differences).

3 Functions

Learning Goals

- Define functions formally as relations.
- Recognize domain, range, injective/surjective/bijective functions.
- Work with function inverses.

Content

- **Relations:** sets of ordered pairs.
- **Cartesian product:** $A \times B$.
- **Functions:** special relations where each input has exactly one output.
  - Notation: $f : A \to B$.
- **Types of functions:** Injective (one-to-one), Surjective (onto), Bijective (both).
- **Inverses:** only for bijective functions.
- **Examples:**

- $f(x) = x$ (identity, bijective).
- $f(x) = x^2$ (not injective over $\mathbb{R}$).
- $f(x) = e^x$ (injective, bijective when restricted).

Exercises

- Identify whether functions are injective/surjective/bijective.
- Find domains and ranges of functions.
- Explore function inverses.

4 Series

Learning Goals

- Use sigma notation.
- Distinguish finite vs. infinite series.
- Test convergence of infinite series.

Content

- **Notation:** $\sum$ and examples.
- **Examples:** geometric series, Leibniz formula for π, Taylor series.
- **Convergence:**
  - Zeno's paradox as motivation.
  - Convergence tests (term test, integral test).
- **Partial sums** and approximations.
- **Products:** $\prod$ notation; connection to sums via logarithms.

Exercises

- Compute partial sums for geometric and harmonic series.
- Use convergence tests on simple infinite series.

5 Floating-Point Arithmetic

Learning Goals

- Explain why computers cannot represent real numbers exactly.
- Understand floating-point representation.
- Identify limitations like machine precision, overflow, underflow.

Content

- **Motivation:** finite precision in computers.
- **Representation:** scientific notation in binary.
- **Conversions:** decimal $\leftrightarrow$ binary.
- **Rounding:** floor, ceil, round.
- **Precision:**
  - ≈ 16 digits in base 10 for double precision.
  - Machine epsilon.
  - Special values: ∞, NaN.
- **Arithmetic behavior:**
  - Addition/subtraction error and loss of significance.
  - Multiplication/division rules for exponents and mantissas.
  - Series truncation when terms fall below machine precision.

Exercises

- Explore floating-point representation of simple fractions in Python.
- Demonstrate loss of precision with $(1 + \varepsilon) - (1 - \varepsilon)$.

Pedagogical Notes

- **Contextualization for Data Science:** show how these foundations connect to later topics like probability, optimization, and algorithms.
- **Python Integration:** use computational demos for truth tables, set membership, floating-point errors.
- **Assessment Ideas:** combine formal proofs (logic/sets) with computational labs (floating-point, series).